

Renjie Gu

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🎓 Education

Fudan University NLP Lab (Graduate Admission Confirmed) 2026.09 – 2029.06
School of Advanced Manufacturing and Intelligent Robotics Computer Science and Technology - Multimodal Large Models and Embodied Intelligence

Central South University 2022.08 – Present
School of Computer Science Computer Science and Technology

- Overall GPA: **3.93/4.0**; Credit-weighted average score: **91.27**; Comprehensive major ranking: **10/268**;
- Selected core course grades: Advanced Mathematics (**97**), Probability Theory (**98**), Computer Organization and Assembly Language (**98**), Discrete Mathematics (**94**), Algorithms (**95**), Principles of Compilers (**94**), Database Systems (**97**), Artificial Intelligence (**94**), Circuit Theory (**95**), College Physics (**97**);

🏢 Publications († denotes co-first authorship)

- [On submission] Renjie Gu, Jiaxu Li, et al. “[Bridging the Detection-to-Abstention Gap in Reasoning Models under Insufficient Information](#)”, NeurIPS 2026
- [On submission] Renjie Gu†, Kaichen Zhou†, Yan Luo, Mengyu Wang. “[GeoWorld-VLM: Geometry from World Models for Vision-Language Models](#)”, NeurIPS 2026 & FMEA @ CVPR 2026 Poster
- [On submission] Yihao Wang, Haoran Xu, Renjie Gu, et al. “[MedMemoryBench: Benchmarking Agent Memory in Personalized Healthcare](#)”, NeurIPS 2026
- [Accepted] Renjie Gu, Jiazhen Du, et al. “[Unlearners Can Lie: Evaluating and Improving “Honesty” in LLM Unlearning](#)”, ACL 2026 Main & NeurIPS 2025 Lock-LLM Workshop
- [Accepted] Boheng Li, Renjie Gu, et al. “[Towards Resilient Safety-driven Unlearning for Diffusion Models against Downstream Fine-tuning](#)”, NeurIPS 2025
- [Accepted] Yishuo Cai†, Renjie Gu†, et al. “[MHALO: Evaluating MLLMs as Fine-grained Hallucination Detectors](#)”, ACL 2025 Findings
- [Accepted] Rongwu Xu, Yishuo Cai, Zhenhong Zhou, Renjie Gu, et al. “[Course-Correction: Safety Alignment Using Synthetic Preferences](#)”, EMNLP 2024

🔧 Research / Internship Experience

Ant Group, Ant Research Institute and Ant Health 2025.12 – 2026.5
Research Intern Advisor: [Yun Yue](#)

- Proposed the Judge-then-solve reasoning paradigm to improve the refusal rate and efficiency of large reasoning models on problems with missing conditions through SFT and reinforcement learning. I implemented the full pipeline, including fine-tuning data construction, reinforcement learning loss function design, and Slime framework debugging. This work has resulted in a first-author paper submitted to NeurIPS 2026.

Harvard University AI and Robotics Lab 2025.8 – 2026.4
MLLM Spatial Reasoning Research Intern Advisor: [Prof. Mengyu Wang](#)

- Proposed a method that aligns the spatial representations from a world model, Lingbot-world-fast, with MLLMs to improve their spatial reasoning ability. This work has resulted in a first-author paper submitted to NeurIPS 2026.
- Tried to improve Rectified Flow by incorporating non-linear velocity, aiming to enhance image generation quality, but it did not work.

Michigan State University OPTML Lab

2025.06 – 2025.10

Trustworthy AI and LLM Unlearning **Summer Intern**

Advisor: [Prof. Sijia Liu](#)

- Participated in the summer research program at the OPTML Lab, focusing on LLM unlearning. I investigated the limitations of existing evaluation methods and improved the forgetting quality of unlearning methods. I proposed an evaluation criterion for honesty in LLM unlearning and further conducted representation-level alignment training to achieve more honest unlearning. This work has resulted in a first-author paper accepted by ACL 2026 Main and the NeurIPS 2025 Lock-LLM Workshop.

Nanyang Technological University

2024.12 – 2025.06

AI Safety and Trustworthy Machine Learning **Research Assistant**

Advisors: [Assoc. Prof. Tianwei Zhang](#),

[Researcher Yiming Li](#)

- Studied concept erasure in text-to-image diffusion models and proposed an unlearning method that is more robust against downstream fine-tuning. I surveyed and summarized existing methods for text-to-image concept erasure, formulated the problem as a bi-level optimization task, and, under the guidance of senior PhD student [Boheng Li](#), drew inspiration from Meta Learning to estimate gradients accurately and efficiently. I implemented the full codebase. This work has been accepted by **NeurIPS 2025**, and I am the second author of the paper.

Research on Multimodal Large Models as Judges for Hallucination Annotation

2024.11 – 2025.03

Advisor: [Prof. Minlie Huang](#)

- Focused on evaluating the fine-grained hallucination detection ability of MLLMs. We constructed a benchmark for evaluating the hallucination annotation ability of models, proposed a metric aligned with human preferences, and investigated how captioning, reasoning, and other test-time scaling methods improve the judging ability of models. Finally, we conducted SFT-based fine-tuning. This work was accepted by **ACL 2025 Findings**.

Research on LLM Safety Alignment Based on Course-Correction

2024.05 – 2025.08

Advisor: [Assoc. Prof. Han Qiu](#)

- Studied safety alignment for LLMs. We found that LLMs exhibit intermediate self-correction when responding to harmful prompts. Based on this phenomenon, we built a benchmark and constructed a dataset, then fine-tuned models using the DPO algorithm to improve safety alignment. This work was accepted by **EMNLP 2024 Industry Track**.

Early Internship Projects and Personal Projects

2023.07 – 2023.09

- During my internship at Beijing Xinxin Xiangqi Technology, I participated in the development of “**Sheila the Kitten**” – an LLM-based iOS companion application: I was mainly responsible for iOS app development, LLM fine-tuning, and agent construction. In addition, I developed many agent-based tools, such as an [AI personal secretary integrated with Notion](#) and a [local voice-interactive AI assistant](#).

♥ Selected Awards

National Outstanding Project, College Student Innovation and Entrepreneurship Training Program 2024.05

National Third Prize, China College Students Service Outsourcing Innovation and Entrepreneurship Competition, Enterprise Track 2024.07

Provincial First Prize, Central South Regional Contest of the China College Students Computer Design Competition 2024.06

National Second Prize, National English Competition for College Students 2022.05

🔧 Selected Honors and Scholarships

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- Scholarships: **Camellia Scholarship** (awarded to only 2 students in the entire school), **First-Class Academic Scholarship of Central South University** (top 5% in the school)
 - Honors: Received the Panchi Star Honor in high school, awarded to only 3 students in the entire grade

Technical Skills

- Programming Languages and Tech Stack: Python (NumPy, Pandas, PyTorch, LLaMA-Factory, Slime), Swift
- Basic Skills: Git, $\text{\LaTeX}$, VS Code, SSH, Codex, Claude Code
- Languages: CET-4 602, CET-6 519, IELTS 7.0

More About Me

- My research interests are broad, including LLM alignment and studies from the perspectives of representations and attention mechanisms. I am also interested in LLM applications such as deep research and MLLM agents. In addition, I am highly interested in topics related to multimodal large models, including MLLM understanding, unified MLLMs, and world models, which will be the research focus of my next stage.
- I have a strong passion for artificial intelligence technologies, including LLMs, MLLMs, and diffusion models. In my spare time, I enjoy reading papers and blogs that interest me. I have independently studied online courses by Hung-yi Lee and Mu Li, as well as Stanford CS231n and CS336. I am also enthusiastic about applying technologies to solve real-world problems, and I have developed my own small projects, such as AI voice assistants, AI secretaries, and iOS apps.
- I enjoy communication, teamwork, and helping others. I am cheerful, open-minded, diligent, and responsible. I love sports, including weight training, badminton, running, and swimming. I also enjoy reading, watching films, and traveling.
- Life philosophy: Across ancient times and vast spaces, I hope to broaden and deepen my life experiences. At the same time, I want to actively participate in AI research, a career that I find meaningful and fascinating and that can provide me with the economic foundation to explore other possibilities, while producing solid work.